

What FIN Determines and What It Does Not

Thirty Adaptive Falsification Studies

Signed kernels, dual dynamics, collective information,
and a conditional operational bridge

Programmes ST8549–ST8578

Krzysztof Żuchowski

Independent Researcher — Fractal Information Theory Project
ORCID: 0009-0002-0909-3613

6 September 2026

One analytical and computational campaign, organized into thirty bounded investigations. Every step states its premises, an explicit result and the reason for the next step. The central result is a conditional uniqueness and stability theorem for collective Markov dynamics, together with an explicit family proving that the presently supplied FIN data do not enforce its decisive premise.

Abstract

The strict finite FIN kernel supports exact spectral calculus, but that calculus does not identify a unique microscopic or physical realization. We test this statement constructively rather than treating it as a universal philosophical prohibition. A positive double cover retains the signs of the canonical legacy profile; its coarse generator is nevertheless not strict. Repeated measurements of the strict Hamiltonian produce squared edge rates, whereas an exact Lindblad heat embedding leaves independent coherence data.

The principal counterexample is a family of collective Markov generators with the same strict singleton dynamics, every singleton path law, product equilibrium, detailed balance, exchangeability, projective consistency and graph symmetries. Their pair records differ. Neither entropy nor consistency alone selects independence. Conversely, autonomous singleton generators and zero simultaneous-pair activity force the independent tensor-sum generator. A nonnegative pair-activity budget also bounds the error in the full finite-time transition law. This supplies an explicit, falsifiable extra premise, not a derivation of that premise from FIN.

Finite resolution, loss and unknown detector calibration are then used to attack the proposed test. A conditional finite-shot protocol survives known resolution-preserving loss; uncalibrated merging can make distinct models exactly indistinguishable. Additional counterexamples show that a static propagator does not determine nonlinear interactions or hidden memory. No physical units, laboratory observations, selector closure, legacy role transfer, Standard Model or gravitational dynamics are obtained.

Contents

1	Executive result and epistemic scope	3
1.1	Repository coverage and limitations	3
2	Founding interpretation and the mathematical inputs	3
2.1	Strict and legacy must remain different typed objects	4
3	Signed legacy: a constructive dilation and its obstruction	4
4	Dual dynamics survives; a unique operational identification does not	5
4.1	Short-time falsification and repeated measurements: rounds 5–7	5
4.2	Exact quantum heat embedding and a coherence ambiguity: rounds 8–10	6
4.3	Imaginary-time filtering: round 11	6
5	The central counterexample: collective dynamics hidden from all singleton laws	6
6	A positive theorem: composition locality, its witness and its stability	7
6.1	Premise removal and the meaning of independence	9
6.2	A deterministic macroscopic limit is too weak: round 18	9
7	From a mathematical witness to a falsifiable counting protocol	9
7.1	Local integer constraints, and a false finite-bin application	9
7.2	Loss that preserves resolution: round 21	10
7.3	A finite-resolution null bound: rounds 22–23	10
7.4	Exact detector aliases: rounds 24–25	11
8	Independent attacks on equilibrium, action and compressed memory	11
8.1	Correcting the parity example without rewriting history: round 26	11
8.2	A propagator does not determine nonlinear interactions: round 27	11
8.3	Identical static compression, different memory: round 28	12
9	What is proved, refuted, conditional and still open	12
10	Literature, originality and the role of the present results	13

11 Recommended research: a ranked local programme	13
12 Verification, reproducibility and final interpretation	15
A Numerical comparison of four decisive distinctions	16
B The complete thirty-step adaptive chain	16

1 Executive result and epistemic scope

The strongest conclusion of this campaign is not that FIN is impossible, and not that it has become a Theory of Everything. It is the following sharpened statement:

Within a precisely declared finite Markov realization class, the present FIN data leave a genuine collective-dynamics ambiguity. A nonnegative, conditionally observable pair-coincidence budget isolates one missing composition premise. Setting that budget to zero yields a unique generator; bounding it yields a quantitative approximation theorem. FIN has not yet supplied the law that sets or bounds it.

This is useful because an unspecified “missing physical principle” becomes a concrete mathematical question. It is only one part of the physics bridge: composition locality is not spatial locality, no-signalling, a light cone, an action unit, a quantum measurement postulate or an actual detector.

We use the categories *Proven*, *Strong evidence*, *Moderate evidence*, *Speculative* and *Refuted*. Here, Proven always means that the displayed mathematical statement follows from its displayed premises by the accompanying proof. Numerical residuals, spectral values and power calculations are floating checks unless explicitly identified as exact rational or symbolic tests. Refuted refers to a stated implication, not to the entire FIN programme. No mathematical priority or independent peer review is asserted.

1.1 Repository coverage and limitations

The initial inventory covered the repository file list and the complete AGENTS section index. The selected reading included the top-level guardrails, current ST8547/ST8548 frontier, relevant earlier kernel, duality, refinement, configuration and source-obstruction sections, SUMMARY_GROK.md and the current theory compendium. The latest committed head at the start was 295cc900; the then-visible scientific inventory, excluding the named third-party directories, contained about 33,000 files. Existing local changes were preserved. No remote pull or heavy algebra campaign was started.

This is *not* a fresh line-by-line verification of every historical file or every assertion in the approximately one-megabyte AGENTS chronology. Historical results were used as a state map; the claims promoted in this report have their own proofs or counterexamples below. In particular, earlier empty generated status payloads and hash matches are not counted as research evidence. Thirty steps here means thirty bounded investigations, not thirty independent papers or thirty universal new theorems.

The campaign proceeded through six checkpoints: signs and vertex dynamics (1–5), measurement constructions (6–10), microscopic consistency (11–15), pair locality and false count tests (16–20), operational identifiability (21–25), and independent action/memory counterexamples plus synthesis (26–30). The results and next-step rationales are retained in `results.json`; the appendix gives the full chain. Replay is deterministic and is not itself a new adaptive discovery run.

2 Founding interpretation and the mathematical inputs

FIN originates in the hypothesis that the nadsoliton itself is primordial, self-coupled fractal information in a persistent solitonic state, with the intended order nadsoliton $\rightarrow$ light $\rightarrow$ matter $\rightarrow$ emergent observer. No separate information substrate is inserted beneath that object. This ontology is the motivation, not an experimentally established conclusion. Likewise, persistent modes, self-learning, maximum resonance, octave labels and fractal compression do not by themselves define a nonlinear soliton equation, physical matter, a metric or a calibrated clock.

2.1 Strict and legacy must remain different typed objects

Let $d(i, j) = \min(|i - j|, 12 - |i - j|)$ and set diagonal weights to zero. The supplied strict profile and the canonical legacy instance used here are

$$W_{ij} = \frac{\cos(0.18575d + 0.16250)}{1 + d^{9/5}}, \quad (1)$$

$$V_{ij} = 4 \log 2 \frac{\cos(\pi d/4 + \pi/6)}{1 + 0.01d}. \quad (2)$$

The latter is also one member of the reconstructed Legacy* class with free amplitude and damping. Statements about this frozen member must not be promoted to every historical legacy variant. The strict phase decimals are supplied frozen parameters, not newly derived constants.

For strict, every angle on $d = 1, \dots, 6$ lies strictly between zero and $\pi/2$. Thus $W_{ij} > 0$ off the diagonal. With

$$A = \text{diag}(W\mathbf{1}) - W = sI - W, \quad s \simeq 1.6603072787660986,$$

the exact identity

$$f^* A f = \frac{1}{2} \sum_{i,j} W_{ij} |f_i - f_j|^2 \geq 0$$

proves positivity. We use row generators $Q = -A$ for stochastic matrices and column probability vectors $\dot{p} = Q^T p = -Ap$; symmetry makes the matrices coincide, not the conventions.

The finite spectral theorem gives

$$U_t = e^{-itA}, \quad P_t = e^{-tA}, \quad G_z = (A + zI)^{-1}, \quad C_t = \cos(t\sqrt{A}).$$

These share spectral projectors. They are respectively a unitary group, Markov heat semigroup, resolvent, and second-order wave evolution factor. U_t has frequencies λ , whereas C_t has $\sqrt{\lambda}$. They do not supply the same operational law, and no SI units are assigned.

3 Signed legacy: a constructive dilation and its obstruction

Theorem 1 (Positive double cover; rounds 1–4). *For any real symmetric zero-diagonal V , put $V_+ = \max(V, 0)$, $V_- = \max(-V, 0)$ entrywise. Define*

$$\widetilde{W} = \begin{pmatrix} V_+ & V_- \\ V_- & V_+ \end{pmatrix}, \quad \widetilde{L} = \text{diag}(\widetilde{W}\mathbf{1}) - \widetilde{W}, \quad E_{\pm} = \frac{1}{\sqrt{2}} \begin{pmatrix} I \\ \pm I \end{pmatrix}.$$

Then

$$\widetilde{L}E_+ = E_+ L_{|V|}, \quad L_{|V|} = \text{diag}(|V|\mathbf{1}) - |V|, \quad (3)$$

$$\widetilde{L}E_- = E_- L_V^{\text{sign}}, \quad L_V^{\text{sign}} = \text{diag}(|V|\mathbf{1}) - V \succeq 0. \quad (4)$$

Mass collapse $C = (I \ I)$ obeys $C\widetilde{L} = L_{|V|}C$.

Proof. Block multiplication gives the intertwining identities. Positivity also follows directly from $f^* L_V^{\text{sign}} f = \sum_{i < j} |V_{ij}| |f_i - \text{sgn}(V_{ij})f_j|^2$. The collapse identity has no dependence on the edge signs. Exponentiating the finite identities gives the corresponding semigroup relations. $\square$

The odd coordinate is a *difference* between sheet populations, not a probability vector. It retains a signed amplitude without interpreting negative numbers as rates. This is standard signed-graph structure [1], applied explicitly here; it is not a new strict kernel.

The raw legacy Laplacian $L_V^{\text{raw}} = \text{diag}(V\mathbf{1}) - V$ satisfies

$$L_V^{\text{sign}} = L_V^{\text{raw}} + 2 \text{diag}(V_- \mathbf{1}).$$

For this circulant instance the shift is a scalar $c \simeq 27.8057289704$. Therefore

$$e^{-itL_V^{\text{sign}}} = e^{-ict} e^{-itL_V^{\text{raw}}}, \quad e^{-tL_V^{\text{sign}}} = e^{-ct} e^{-tL_V^{\text{raw}}}.$$

The first factor is a global unitary phase; the second is a real attenuation. Removing it is not a probability-preserving gauge change. This is another reason to keep the operational meanings of the dual channels distinct.

Proposition 2 (Frustration obstruction). *The canonical legacy instance cannot be switched to all-positive weights by vertex signs, nor made into positive rates by a positive diagonal similarity transformation.*

Proof. In triangle $(0, 1, 2)$ the two length-one edges are positive and the length-two edge is negative: $\cos(5\pi/12) > 0$ and $\cos(2\pi/3) < 0$. The product of edge signs around a cycle is unchanged under $V_{ij} \mapsto \sigma_i V_{ij} \sigma_j$. An all-positive triangle would have positive product. A positive similarity multiplies each edge by a positive ratio and cannot change any sign in the first place. $\square$

The computed legacy signed-sector lowest eigenvalue is about 6.42867; the raw Laplacian has a negative eigenvalue about -21.3771 . These numbers are checks, while the sign obstruction is analytic. There are 112 negative-product triangles in the finite enumeration. The cover forgets all signs under mass collapse and produces $|V|$, not W . Even the best scalar multiple of $|V|$ has relative Frobenius error about 0.8834 against strict. Exact nonproportionality already follows from unequal ratios at two distances. No amplitude fit is used as a source.

This construction does not complete legacy into strict, select a physical sign, explain electroweak or gravitational roles, or supply a fractal $12 \rightarrow 24$ refinement law. Its extra sheet is a mathematical sign carrier.

4 Dual dynamics survives; a unique operational identification does not

4.1 Short-time falsification and repeated measurements: rounds 5–7

For $i \neq j$, Taylor expansion gives

$$(P_t)_{ij} = tW_{ij} + O(t^2), \quad |(e^{-itH})_{ij}|^2 = t^2|H_{ij}|^2 + O(t^3).$$

For nonzero W no positive linear time calibration identifies these transition records near zero. This does not refute common spectral calculus; it refutes equality of these specified vertex-record laws.

Proposition 3 (Measured small-step limit). *If each step of duration δ applies $e^{-i\sqrt{\delta}H}$ followed by a complete vertex measurement, its stochastic matrix S_δ obeys*

$$S_\delta = I + \delta Q_H + O(\delta^{3/2}), \quad (Q_H)_{ij} = |H_{ij}|^2 \quad (i \neq j).$$

For fixed T , $S_{T/n}^n \rightarrow e^{TQ_H}$. Without the $\delta^{-1/2}$ Hamiltonian scaling, the same frequent-measurement limit freezes the vertex transitions.

Proof. Expand the unitary to second order in $\sqrt{\delta}$. Off-diagonal probabilities are $\delta|H_{ij}|^2 + O(\delta^{3/2})$; stochasticity fixes the diagonal. The finite matrix product limit follows from the logarithm near the identity, or a telescoping estimate with error $O(n^{-1/2})$. With an unscaled Hamiltonian the off-diagonal step probability is $O(\delta^2)$ and $n\delta^2 \rightarrow 0$. $\square$

For $H = A$ the limiting rates are W_{ij}^2 . Since the six strict weights differ, no common clock makes those rates equal to W_{ij} . One can *compile* the desired limit by choosing

$$H_{ij} = \sqrt{W_{ij}} e^{i\theta_{ij}}, \quad \theta_{ji} = -\theta_{ij}, \quad H_{ii} \in \mathbb{R}.$$

This is a construction with new phase and diagonal data. It is not $H = A$. Changing one real edge sign preserves all compiled rates but changes a triangle phase and $\text{Tr}(H^3)$; hence even its coherent spectrum need not be preserved. Gauge-invariant loop phases survive diagonal rephasing.

4.2 Exact quantum heat embedding and a coherence ambiguity: rounds 8–10

Use the rank-one jumps $L_{ji} = \sqrt{W_{ij}}|j\rangle\langle i|$ and set

$$\mathcal{D}_W(\rho) = \sum_{i \neq j} \left[L_{ji} \rho L_{ji}^* - \text{frac}12 \{ L_{ji}^* L_{ji}, \rho \} \right], \quad (5)$$

$$\mathcal{D}_W(\rho)_{ii} = (Q^T p)_i, \quad p_i = \rho_{ii}, \quad (6)$$

$$\mathcal{D}_W(\rho)_{ij} = -\frac{1}{2}(r_i + r_j)\rho_{ij} \quad (i \neq j), \quad r_i = \sum_j W_{ij}. \quad (7)$$

The displayed Lindblad form proves complete positivity and trace preservation of its finite-dimensional semigroup [2]. The component identities follow by direct multiplication and hold for *every* density matrix, including coherent states. Thus exact heat populations admit a quantum extension. This is within the established quantum-stochastic-walk framework [3].

For every $\gamma \geq 0$ the additional vertex-dephasing generator

$$\gamma \left(\sum_i |i\rangle\langle i| \rho |i\rangle\langle i| - \rho \right)$$

changes off-diagonal decay but none of those population trajectories. Complete population records do not identify γ .

Proposition 4 (Restricted Hamiltonian obstruction). *In the class consisting of these rank-one jumps, arbitrary vertex dephasing and a Hamiltonian commutator, autonomous populations $\dot{p} = Q^T p$ for every density matrix require the Hamiltonian to be diagonal in the vertex basis.*

Proof. The jump and dephasing terms already give the required diagonal. The remaining condition is $\text{diag}[-i(H\rho - \rho H)] = 0$ for every density matrix. For each edge, vary the phase of a small coherence in a positive two-dimensional block. Its contribution is proportional to the imaginary part of $H_{ij}\rho_{ji}$. Vanishing for every phase forces $H_{ij} = 0$. Conversely a diagonal Hamiltonian contributes no diagonal commutator. $\square$

This excludes adding the non-diagonal strict $H = A$ in this particular embedding while retaining exact heat for all states. It is not a theorem about every conceivable Lindblad representation, dilation, feedback instrument or restricted preparation class.

4.3 Imaginary-time filtering: round 11

For a positive contraction $P = e^{-tA}$ the operation $\rho \mapsto P\rho P$ is completely positive but generally trace decreasing. Its normalized conditional state is

$$\rho \mapsto \frac{P\rho P}{\text{Tr}(P\rho P)}.$$

If P has two unequal eigenvalues, mixing the corresponding pure states before filtering changes their relative weights, while filtering each pure state first leaves it unchanged. Thus the normalized map is not affine and cannot be a deterministic quantum channel on every input. A successful postselection event and its probability must be retained. The analytic identity $P_t = U_{-it}$ has not produced a Born rule or a measurement record.

5 The central counterexample: collective dynamics hidden from all singleton laws

Let $X = \{0, \dots, 11\}$ and declare a finite population state space X^N . This is a mathematical completion, not a derivation of physical particles or of the earlier 4096-state binary configuration

model. For $x = (x_1, \dots, x_N)$ define

$$(G_N^{\text{ind}} f)(x) = \sum_{k=1}^N \sum_{j \neq x_k} W_{x_k j} [f(x^{k \rightarrow j}) - f(x)], \quad (8)$$

$$(G_N^{\text{com}} f)(x) = \sum_{i < j} W_{ij} [f(\sigma_{ij} x) - f(x)], \quad (9)$$

$$G_N^{(\theta)} = (1 - \theta)G_N^{\text{ind}} + \theta G_N^{\text{com}}, \quad 0 \leq \theta < 1, \quad (10)$$

where σ_{ij} exchanges vertex labels i, j in *every* coordinate at once. Terms that do not change x are zero.

Theorem 5 (Explicit collective nonidentifiability; rounds 12–14 and 29). *Every member of this family has the same autonomous singleton generator $Q = -A$ and hence every singleton path law for any supplied singleton initial distribution. The family is exchangeable under particle-label permutations and projectively consistent under deletion of coordinates. Each finite N member is irreducible and reversible with respect to the same product-uniform measure. It respects every vertex symmetry of W . For $N \geq 2$, different θ nevertheless give different collective transition laws.*

Proof. For $f(x) = h(x_k)$, the only common transpositions that affect f exchange x_k with some j , giving precisely $\sum_j W_{x_k j} [h(j) - h(x_k)]$. The independent generator gives the same result. The calculation extends to any subset of coordinates: jumps on deleted independent coordinates vanish, and the common transpositions restrict to the same common-transposition generator on the retained ones. This proves strong projective consistency and singleton path equality.

Coordinate permutations commute with both constructions. Vertex symmetries permute transpositions and preserve their weights. Each common map is an involution with the same forward and reverse coefficient; the independent weights are symmetric as well. Their generator matrices are symmetric, so the uniform measure on X^N is reversible. For $\theta < 1$, the positive independent part connects every configuration by single-coordinate changes, proving irreducibility and uniqueness of this stationary distribution.

At $N = 2$, the rate from (i, i) to (j, j) is θW_{ij} . It is zero for the independent generator and positive for $\theta > 0$. For different parameters these entries differ. Therefore the pair semigroups, and consequently the collective path laws, differ. $\square$

The construction is compatible with the general literature on exchangeable Markov processes, where exchangeability permits shared randomness rather than forcing unconditional independence [4]. The proof above is finite and does not require the general representation theorem.

All members have the same stationary occupancy multinomial distribution and the same static Shannon large-deviation expression. Thus even the full product equilibrium law, rather than only its entropy, fails to identify the kinetics. If initialized in that equilibrium, the coordinates are independent at each *single instant*; their path histories need not be independent. This distinction prevents an erroneous inference from static factorization to dynamical tensorization.

For strict W , preparation $(0, 0)$ and time $t = 0.02$, direct exponentiation gives

$$\mathbb{P}_{\theta=0}((1, 1)) \simeq 0.0000834915, \quad \mathbb{P}_{\theta=0.35}((1, 1)) \simeq 0.0031556561.$$

The numerical values are illustrations. The exact differing first-order rates already prove distinguishability. No laboratory observation was made.

6 A positive theorem: composition locality, its witness and its stability

Consider *any* finite continuous-time Markov generator G on X^N , with off-diagonal rates $g(x, y) \geq 0$. The pointwise autonomous-singleton condition is

$$\sum_{y: y_k = j} g(x, y) = Q_{x_k j} \quad (j \neq x_k), \quad \text{for every } x, k, j. \quad (11)$$

This is stronger than matching a mean at equilibrium or along one chosen trajectory. Define the number of changed coordinates in one event and its pair-activity budget by

$$K(x, y) = \#\{k : x_k \neq y_k\}, \quad B(x) = \sum_{y \neq x} g(x, y) \binom{K(x, y)}{2}.$$

Theorem 6 (Conditional uniqueness; rounds 15–16). *Under (11), $B(x) = 0$ for every configuration if and only if $G = \sum_k Q^{(k)}$, the independent transition generator. No exchangeability premise is required.*

Proof. All summands in B are nonnegative, so $B = 0$ excludes every event changing two or more coordinates. For a fixed x, k, j , exactly one possible remaining destination changes only coordinate k to j . Equation (11) therefore forces its rate to be $Q_{x_k j}$. The diagonals are fixed by row sums, proving the claimed generator. Conversely this generator changes one coordinate per event, hence has zero budget. $\square$

For a fixed initial configuration x , define $f_k^x(z) = \mathbf{1}_{z_k \neq x_k}$. Then

$$\Gamma_G(f_k^x, f_l^x)(x) := G(f_k^x f_l^x)(x) - f_k^x(x) G f_l^x(x) - f_l^x(x) G f_k^x(x) = \sum_y g(x, y) \mathbf{1}_{y_k \neq x_k, y_l \neq x_l}.$$

Summing over $k < l$ gives $B(x)$. This uses gross changes in an event, not cancellations of signed displacements. Merely finding zero *stationary average* cross covariance for one pair of fixed observables is insufficient: in the common-flip counterexample such an average can vanish while $B > 0$. That false shortcut is explicitly rejected by a regression test.

Theorem 7 (Finite-time robustness; round 17). *Under (11), if $\sup_x B(x) \leq \varepsilon$, then for every initial configuration and $t \geq 0$,*

$$\text{TV}(e^{tG}(x, \cdot), e^{tG^{\text{ind}}}(x, \cdot)) \leq \min(1, 2\varepsilon t).$$

The same bound holds after mixing initial configurations with any common initial distribution and after any common stochastic observation map.

Proof. In one row, a joint event $x \rightarrow y$ with rate c and $k = K(x, y) \geq 2$ replaces k separate single-coordinate events of rate c each, by the singleton constraints. Its contribution to $G - G^{\text{ind}}$ is

$$c \left[\delta_y - \sum_{l \text{ changed}} \delta_{x^l \rightarrow y_l} + (k-1)\delta_x \right].$$

Its row ℓ^1 norm is at most $2kc$. Consequently

$$\frac{1}{2} \sup_x \sum_y |G - G^{\text{ind}}|(x, y) \leq \sup_x \sum_{y: K \geq 2} K g(x, y) \leq 2 \sup_x B(x),$$

because $k \leq 2 \binom{k}{2}$. Apply the finite matrix Duhamel identity

$$e^{tG} - e^{tG^{\text{ind}}} = \int_0^t e^{sG} (G - G^{\text{ind}}) e^{(t-s)G^{\text{ind}}} ds.$$

Stochastic multiplication contracts the relevant row total-variation norm. Integration gives $2\varepsilon t$; total variation is at most one. Mixtures and observation channels also contract total variation. $\square$

This bounds *transition distributions*, not the total variation of entire continuous-time path records. Pathwise coupling bounds require their own accounting. The stated bound is conservative, not claimed sharp. The three-coordinate test has actual transition TV about 0.04513, below its 0.672 bound; twelve further perturbed three-state tests passed.

6.1 Premise removal and the meaning of independence

Removing zero pair activity restores the common-transposition family. Removing the singleton-rate condition permits, for example, $2 \sum_k Q^{(k)}$ with the same zero budget. Thus the two conditions are independently needed for this characterization relative to the declared finite Markov state space. A non-Markov model, unidentified population or different state space is outside the theorem, not silently covered.

A product transition semigroup also does not force a product *initial* law. Initially correlated particles remain correlated at finite times. The model specification must supply that law separately. The theorem selects a generator, not the origin of particles, their preparation, a physical clock or an apparatus.

6.2 A deterministic macroscopic limit is too weak: round 18

For each N , select unordered pairs of particle labels. For every vertex transposition (i, j) , let it act on each selected pair at rate $\theta W_{ij}/(N-1)$, together with independent jumps at rate $(1-\theta)W_{ij}$ per coordinate. Each coordinate participates in $N-1$ pairs and still has precisely the generator Q . The family is exchangeable, but it is not projectively consistent across N : deleting a partner changes some paired events into singleton events.

In the two-state case, with all particles initially at one vertex and flip rate w , the total numbers of changed coordinates have initial rates

$$r_1 = (1-\theta)Nw, \quad r_2 = \theta Nw/2, \quad \mu/N = w, \quad v/N = (1+\theta)w.$$

The empirical fraction has jumps of size at most $2/N$, total event rate $O(N)$, and hence martingale quadratic variation $O(1/N)$ on bounded time intervals. Its drift is the exact linear heat drift. With deterministic or appropriately concentrated initial data, the usual elementary martingale estimate and variation of constants yield a deterministic mean-field limit. Nevertheless the local noise coefficient differs. We do not claim a full central-limit theorem here. Exact projectivity, a deterministic limit and absence of common jumps are different assumptions.

7 From a mathematical witness to a falsifiable counting protocol

7.1 Local integer constraints, and a false finite-bin application

For an integer-mark event process with instantaneous nonnegative rates c_k , its local counting Hamiltonian is $h(s) = \sum_k c_k (e^{ks} - 1)$. Assuming the displayed moments are finite,

$$\kappa_2 = \sum_k k^2 c_k \geq |\sum_k k c_k| = |\kappa_1|, \quad \kappa_4 - \kappa_2 = \sum_k k^2 (k^2 - 1) c_k \geq 0.$$

Equality in the second inequality forces all nonzero marks to have $|k| = 1$. These statements concern instantaneous raw count rates, not the finite-window cumulants of an arbitrary correlated record.

For the ST8548 strict edge test, the quadratic logarithmic-mean noise prescription has $v/|\mu| = 2/\log 12 < 1$. This refutes its exact interpretation as such an integer-count generator. It does not refute a continuous Gaussian approximation or the valid Onsager gradient identity. The nonquadratic microscopic fluctuation construction is established literature [6]; its FIN application is not a new universal theory.

Round 20 deliberately breaks an attempted experimental shortcut. Poisson arrivals with rate λ , reported as at most one click in a bin of width Δ , produce a Bernoulli variable with

$$q = 1 - e^{-\lambda\Delta}, \quad \frac{\text{Var}(\text{click})}{\mathbb{E}(\text{click})} = e^{-\lambda\Delta} < 1.$$

Valid microscopic jumps can therefore yield a sub-unit ratio after ordinary thresholding. That observed ratio alone is not a falsification of Markov microscopy or evidence for a novel FIN noise law.

7.2 Loss that preserves resolution: round 21

If each changed coordinate in an event is independently registered with efficiency $0 < \eta \leq 1$, and the instrument retains the number of changes, then $M \mid K \sim \text{Bin}(K, \eta)$. Counting pairs gives exactly

$$\mathbb{E} \binom{M}{2} = \eta^2 \binom{K}{2}, \quad B_{\text{observed}}(x) = \eta^2 B(x).$$

Thus the ideal zero set survives positive independent loss. An upper bound on B needs a known lower bound on efficiency. Correlated losses, unknown merging, afterpulses, backgrounds or unobserved detector recovery states are not licensed by this identity. It is an event-level identity, not an uncorrected finite-bin factorial-moment formula.

7.3 A finite-resolution null bound: rounds 22–23

Suppose the finite Markov chain has an escape-rate bound Λ . Under the no-joint-change null, at least two detected coordinate changes require at least two true events. Uniformization therefore gives

$$\mathbb{P}(C_\Delta \geq 2) \leq \mathbb{P}(\text{Pois}(\Lambda\Delta) \geq 2) \leq \frac{(\Lambda\Delta)^2}{2} =: q_0. \quad (12)$$

The last bound follows from $\mathbf{1}_{M \geq 2} \leq M(M-1)/2$ and the Poisson factorial moment. For the strict N -particle autonomous-marginal class, $\Lambda = Ns$ is valid: every genuine event changes at least one coordinate and the sum of coordinate change rates is Ns .

If a prepared configuration x instead has joint-event detected rate

$$r_{\geq 2}^{\text{det}}(x) = \sum_y g(x, y) \mathbb{P}(\text{Bin}(K(x, y), \eta) \geq 2),$$

the probability of exactly one uniformization proposal, realizing that event with no others, gives the lower bound

$$\mathbb{P}_x(C_\Delta \geq 2) \geq e^{-\Lambda\Delta} \Delta r_{\geq 2}^{\text{det}}(x).$$

The null is $O(\Delta^2)$; a resolved common-event alternative is $O(\Delta)$. Neither result requires approximating a finite closed-chain count by independent Poisson edge counts.

For a completely explicit design example, take two binary coordinates, flip rate $w = 0.7$, common-flip fraction $\theta = 0.2$, efficiency $\eta = 0.6$ and $\Delta = 0.01$. The independent and common Poisson clocks have configuration-independent rates in this example. After thinning, the observed count law has marks one and two with rates

$$c_1 = 2w\eta(1 - \theta\eta), \quad c_2 = \theta w\eta^2, \quad h(s) = c_1(e^s - 1) + c_2(e^{2s} - 1).$$

Hence

$$q_1 = 1 - e^{-(c_1+c_2)\Delta}(1 + c_1\Delta) \simeq 0.00053104588, \quad q_0 \leq 0.000098.$$

With 200,000 independently reset trials, the recommended conservative rule rejects when at least 40 bins have two or more detected changes. This rule admits error bounds without relying on a floating binomial tail. Under the null, for the count Z of qualifying bins,

$$\mathbb{P}(Z \geq 40) \leq 2^{-40}(1 + q_0)^M \leq 2^{-40}e^{19.6} < (11/16)^{20} < 10^{-3},$$

using $e < 11/4$. Under the supplied alternative, $q_1 \geq (1 - 0.014)0.010.20.70.6^2 = 0.000496944$, so $Mq_1/2 > 49$. Using $e > 8/3$ gives

$$\mathbb{P}(Z \leq 39) \leq 2^{39}(1 - q_1/2)^M \leq 2^{39}e^{-Mq_1/2} < \frac{3^{49}}{2^{108}} < 10^{-9}.$$

The simple inequalities for e follow from its convergent positive Taylor series and a geometric tail bound. The final bounds are exact rationals, checked with integer arithmetic. Floating binomial tails supply an additional check; their less conservative threshold 36 is not needed for the certified rule. These are conditional design calculations, not completed experiments, detector specifications or evidence that physical FIN constituents exist.

7.4 Exact detector aliases: rounds 24–25

Two attacks survive against less restrictive instruments.

First, in the binary two-particle model the event rate is $(2 - \theta)w$. An instrument reporting one click per event, regardless of whether one or two coordinates change, sees a Poisson process at that rate. Independent single flips, thinned with efficiency $(2 - \theta)/2$, give exactly the same nonzero Poisson record law. Both microscopic models can be irreducible. This is a nondegenerate alias, not a detector that always returns nothing.

Second, even retaining mark counts leaves a calibration ambiguity. The parameter triples

$$(w, \theta, \eta) = (0.7, 0.2, 0.6), \quad (1.4, 0.4, 0.3)$$

have identical c_1, c_2 and therefore identical *entire* counting process laws. More generally the observed mean rate μ and double-mark rate b give only

$$w = \frac{\mu}{2\eta}, \quad \theta = \frac{2b}{\mu\eta},$$

with η still free in its admissible interval. Higher count cumulants cannot recover an extra parameter when the full likelihood already agrees. This concerns this count-only protocol; calibrated singleton trajectories, efficiency or another independent operational datum can break the alias.

8 Independent attacks on equilibrium, action and compressed memory

8.1 Correcting the parity example without rewriting history: round 26

The prior ST8548 audit correctly refuted an earlier claim about the energy $E(x) = -\vartheta x_1 x_2 x_3$, $x_i \in \{-1, 1\}$. Every spin flip has $|\Delta E| = 2|\vartheta|$, so

$$Q_{\text{Met}} = (1 + e^{-2|\vartheta|})Q_{\text{HB}}.$$

That example cannot distinguish kinetics beyond an overall clock. Here we retain the correction and add an *explicit* new field:

$$E_h(x) = -\vartheta x_1 x_2 x_3 - h(x_1 + x_2 + x_3).$$

It preserves permutation symmetry among the three labels. A flip now has magnitude $2|\vartheta + h|$ or $2|\vartheta - h|$. The corresponding Metropolis/heat-bath ratios are different for generic nonzero h . At $\vartheta = 0.3, h = 0.17$ they are about 1.3906278354 and 1.7710515858. Both laws still have the same Gibbs equilibrium because their forward/reverse ratio is $e^{-\Delta E}$. Thus this enlarged example really is a non-clock kinetic counterexample. The field is an additional model parameter, not a strict-derived source or a new prism closure.

8.2 A propagator does not determine nonlinear interactions: round 27

Proposition 8 (Inverse-action ambiguity). *Let $M = A + m^2 I \succ 0$. For every $\lambda \geq 0$ the action*

$$S_\lambda(\phi; J) = \frac{1}{2}\phi^T M \phi - J^T \phi + \lambda \sum_i \phi_i^4$$

has a unique stationary point, zero-source Hessian M , and zero-source linear response $G = M^{-1}$. Its nonlinear response depends on λ .

Proof. The Hessian is $M + 12\lambda \text{diag}(\phi_i^2) \succ 0$ and the action is coercive, proving existence and uniqueness of the minimizer. The stationary equation is $M\phi + 4\lambda\phi^{\circ 3} = J$. The implicit-function theorem at $J = 0$ gives linear response G . For $J = \epsilon J_0$, coefficient comparison yields

$$\phi(\epsilon) = \epsilon G J_0 - 4\lambda \epsilon^3 G[(G J_0)^{\circ 3}] + O(\epsilon^5).$$

The cubic coefficient is different for distinct λ and a nonzero J_0 . The coincident propagator has not selected that coefficient. $\square$

This concerns a stationary classical source response or zero-field Hessian. It does *not* assert equality of the fully interacting quantum or thermal two-point function after integrating fluctuations. The mass parameter here is a positive dimensionless regularizer, not a derived physical mass. Reconstructing the smallest quadratic action from a propagator is valid in the declared Gaussian class; selecting that class and discarding all higher interactions is additional information.

8.3 Identical static compression, different memory: round 28

For $a, d > 0$ and real c , consider the positive matrix

$$M(c, d) = \begin{pmatrix} a + c^2/d & c \\ c & d \end{pmatrix}.$$

Its Schur complement at zero is a and $(M^{-1})_{11} = 1/a$, for every c, d . At a frequency variable z , however, the Schur complement of $M + zI$ is

$$S(z) = a + z + \frac{c^2}{d} - \frac{c^2}{d+z}, \quad S'(0) = 1 + \frac{c^2}{d^2}.$$

The corresponding first-order two-variable relaxation, after eliminating the hidden variable, has a kernel $c^2 e^{-dt}$ and a separate term depending on the hidden initial state. Different positive c, d therefore yield identical coarse static Green data and different dynamical memory. For $(c, d) = (1, 1)$ and $(2, 3)$ the slopes are 2 and 13/9.

This proof is an exact two-dimensional counterexample, not a numerical failure to infer a large hidden system. It shows why static Green–Schur compression and a finite self-similar picture cannot by themselves select a unique memory hierarchy. It does not exclude an additional, explicitly derived self-similar composition law that constrains those hidden entries.

9 What is proved, refuted, conditional and still open

Status	Result and exact boundary
Proven	Positive signed double-cover decomposition and frustration obstruction. It retains legacy signs but does not complete legacy into strict.
Proven	Strict unitary and heat functions share a generator. Their specified vertex-record laws have different short-time orders.
Proven, conditional	Measurement-limit and Lindblad constructions realize declared heat laws after supplying instruments or jumps; coherent phases, dephasing and calibration remain independent.
Refuted implication	Singleton FIN dynamics + projective consistency + exchangeability + reversible product equilibrium implies one collective law.

Status	Result and exact boundary
Proven, conditional	Pointwise autonomous singleton rates + zero pair activity uniquely determines the finite Markov generator. A small uniform budget yields the stated transition-TV bound.
Refuted implication	Sub-unit variance/mean in finite detector bins alone refutes a microscopic integer-jump model.
Proven, conditional	Resolution-preserving thinning and an escape bound admit a finite-bin null test. The conservative design has exact rational error bounds as well as floating checks.
Refuted implication	Complete count likelihood without sufficient detector and clock calibration always identifies the microscopic parameter.
Refuted implication	A zero-source propagator uniquely determines nonlinear action or hidden memory.
Open	A strict-sourced composition law constraining pair activity; legacy-to-strict completion and role transfer; internal selector; coherent phase source; physical calibration and apparatus; continuum and physical field content.
Speculative	FIN as the fundamental generative theory of the observed universe. No result in this campaign promotes that ontology to physics.

The explicit counterexamples are a class-scoped obstruction, not a proof that *no* missing theorem can ever exist. A future theorem might use a new source object not shared by the examples. Its value would be measured by which explicit pair of currently admissible completions it excludes, and whether its premises come from FIN rather than from the desired answer.

10 Literature, originality and the role of the present results

Signed switching, balance and signed graph operators are established mathematics [1]. Lindblad generators and quantum stochastic walks already unify specified coherent and incoherent evolutions in a larger conditional framework [2, 3]. The presence of both e^{-itA} and e^{-tA} is consequently not, on its own, a new physical unification theorem.

Exchangeable Markov systems permit common random influences [4]; exchangeability must not be confused with path independence. The explicit weighted-transposition family here makes that distinction transparent for the frozen FIN operator and is directly executable. Logarithmic-mean entropy geometry [5] and nonquadratic Markov fluctuation actions [6] describe supplied microscopic dynamics; they do not source that microscopy from a finite spectrum.

The present contribution is an integrated FIN-specific audit, explicit completion counterexamples, an elementary pair-budget characterization and stability estimate, and a detector-aware falsification design. We do not claim that the elementary generator characterization, matrix dilation, perturbation estimate or thinning identities are first discoveries in mathematics. Nor has this campaign exhausted all modern mathematical literature. The proofs in this report stand on their displayed arguments, not on analogies or the prestige of the cited theories.

11 Recommended research: a ranked local programme

No defensible numerical probability can be assigned to eventual ToE success from this evidence. The ranking below is by immediate information value, specificity of the falsification test and tractability on a local machine, not a percentage forecast.

No.	Next question	Required result / stop rule
1	Strict source of composition locality	Translate the actual coupling law into a generator or action on a declared composite state space. Prove $B = 0$, bound it, or exhibit $B > 0$. Stop if zero activity is merely inserted as independence.
2	Coupling-diagram source audit	Type every positive, negative and mirror coupling as an event, phase or deterministic term. Decide whether it predicts common events. A verbal sign of “information gain” is not a rate.
3	General local completion class	Classify generators with the same pointwise marginals and graph symmetries at $N = 2, 3$. Use exact linear constraints and nonnegative rates; do not infer closure from a sampled rank.
4	Sharp robust budget certificate	Optimize the $2\epsilon t$ bound, including state-dependent preparation classes; derive a separate path-law bound rather than relabelling the transition bound.
5	Minimal calibrated coincidence record	Find the smallest identifiable tuple of efficiency, clock, population, reset and mark resolution. Use likelihood equalities to reject insufficient tuples.
6	Hidden detector dynamics	Add a finite recovery/afterpulse state and derive finite-bin null bounds. Reject any design whose nuisance class reproduces the alternative exactly.
7	Source of coherent loop phases	Test an actual strict coupling law against the square-root-rate Hamiltonian family. Require a phase-sensitive invariant, not another fit to populations.
8	Exact simultaneous quantum closure	Extend the restricted Lindblad analysis to a declared larger class, with complete-positivity constraints and precise preparation/effect scope. Do not extrapolate the restricted no-go.
9	Nonlinear response source	Determine whether a supplied FIN law fixes the cubic response that distinguishes the S_λ family. Keep classical response and interacting quantum correlators separate.
10	Frequency-resolved compression	Seek a refinement composition law that fixes $S(z)$, not just $S(0)$. Export which hidden moments are sourced and which remain observationally invisible.
11	Signed-cover completion test	Ask whether any target-independent, positive operational map from the signed cover yields the strict profile. Failure is evidence only for the declared map class. No legacy role transfer.
12	Exact proof packaging	Formalize the finite generator/budget theorem and its rational removal witnesses; interval-certify any numerical boundary that is actually used in a scientific decision.

The first item has the greatest immediate chance of resolving a *specific* foundational ambiguity because it targets an explicit free parameter family while remaining inexpensive and locally falsifiable. It would still not prove the Standard Model, gravity, physical spacetime or the relation to experiments. A theory of actual physics eventually requires empirical confrontation; no quantity

of local simulation is an independent laboratory record. Until then, predictive conditional models and rigorous obstructions are the honest deliverables.

12 Verification, reproducibility and final interpretation

The package contains the self-contained analytical computation, 33 scientific regression tests, deterministic results for all thirty steps, the source and generated PDF of this report, a numerical figure and a provenance manifest. The checks include exact integer double-cover identities, rational thinning and event-rate identities, symbolic response coefficients, finite matrix exponentials, parameter perturbations and premise-removal counterexamples. Numerical evidence was not promoted to an interval certificate or a proof-assistant result. Random stress cases use seed 8549 and are robustness checks, not exhaustive proofs.

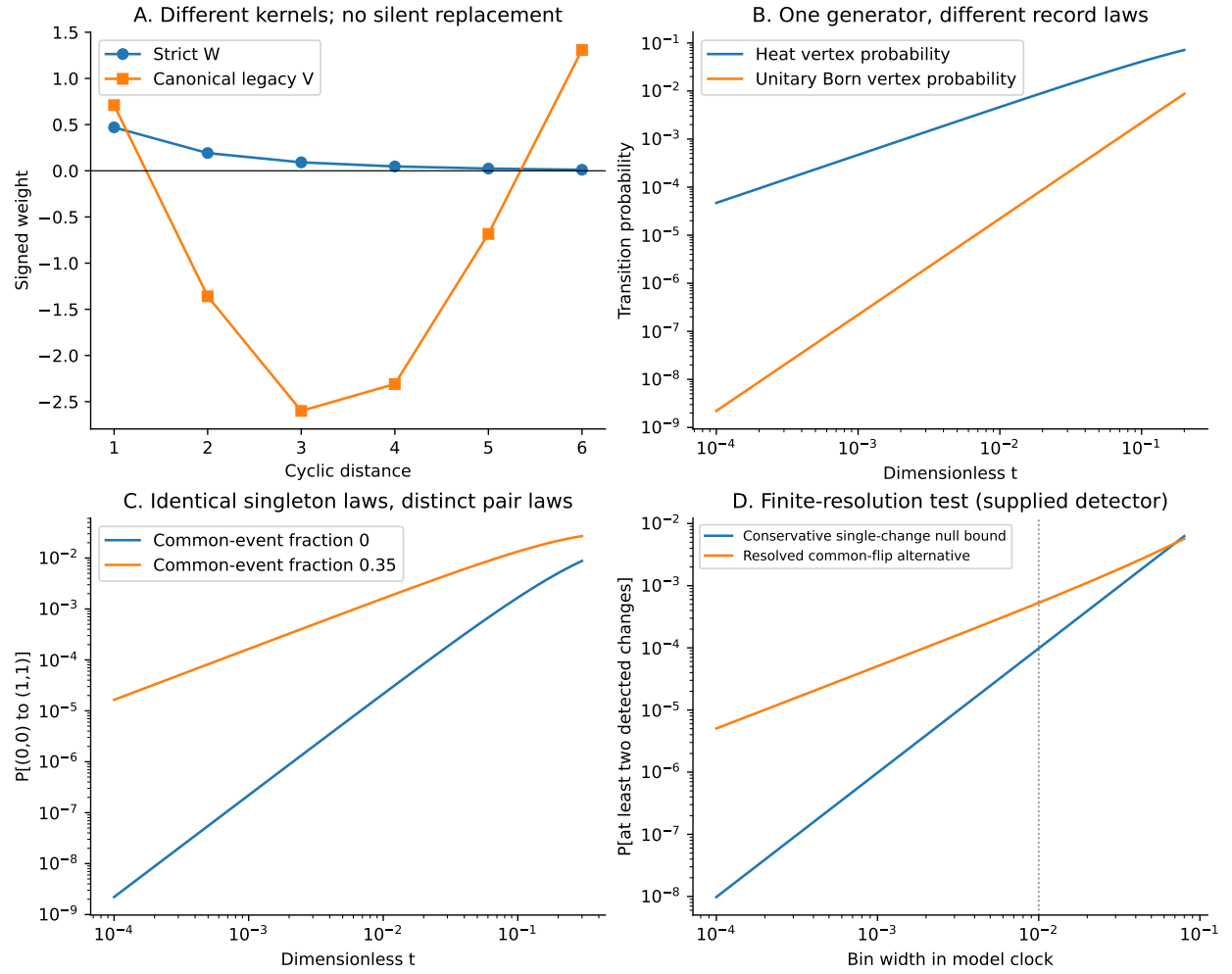
The preceding ST8547 and ST8548 suites are also replayed as compatibility checks; unchanged historical archives remain unchanged. Code and output hashes establish identity of artifacts, not scientific truth. No new DOI is assigned, no old reserved DOI is reused, and nothing is uploaded to Zenodo.

Deepest interpretation that survives this campaign. FIN currently provides a finite spectral-information structure with valid, distinct dynamical calculi and useful conditional completion mechanisms. It does not yet provide one unavoidable physical realization. The most productive missing object exposed here is an internally sourced law of composition: a law fixing which collective changes, coherence relations and hidden responses are actually allowed. For one precise Markov sector, its key obligation is now expressible by a nonnegative pair-activity budget, with both an exact uniqueness theorem and a quantitative error bound. That is a research target with a falsification criterion, not new physics introduced by assumption.

References

- [1] T. Zaslavsky, *Signed graphs*, Discrete Applied Mathematics 4 (1982), 47–74. Original publication.
- [2] G. Lindblad, *On the generators of quantum dynamical semigroups*, Communications in Mathematical Physics 48 (1976), 119–130. Original publication.
- [3] C. A. Rodríguez-Rosario, J. D. Whitfield and A. Aspuru-Guzik, *Quantum stochastic walks: A generalization of classical random walks and quantum walks*, arXiv:0905.2942; Phys. Rev. A 81, 022323 (2010). Author preprint.
- [4] H. Crane and S. P. Lalley, *Exchangeable Markov Processes on $[k]^{\mathbb{N}}$ with Cadlag Sample Paths*, arXiv:1307.1713 (2013). Author preprint.
- [5] J. Maas, *Gradient flows of the entropy for finite Markov chains*, arXiv:1102.5238 (2011). Author preprint.
- [6] A. Mielke, M. A. Peletier and D. R. M. Renger, *On the relation between gradient flows and the large-deviation principle, with applications to Markov chains and diffusion*, arXiv:1312.7591 (2013). Author preprint.

A Numerical comparison of four decisive distinctions



The figure uses only the declared mathematical models. It shows neither experimental data nor a simulation of the universe.

B The complete thirty-step adaptive chain

01. ST8549 – Can both raw kernels be Markov conductances?

Outcome. Strict is positive; the declared raw Legacy* profile is signed and fails the rate test.

Status. Proven (conditional; see proof)

Next-step rationale. Instead of taking absolute values silently, retain signs in a doubled state space.

02. ST8550 – Can legacy signs be retained in a positive stochastic cover?

Outcome. A 24-state positive double cover has unsigned-even and signed-odd Laplacian sectors.

Status. Proven (conditional; see proof)

Next-step rationale. A shifted signed amplitude is not a probability: test whether a positive gauge removes the distinction.

03. ST8551 – Can a diagonal sign or positive Doob gauge make raw legacy positive?

Outcome. A negative triangle product obstructs sign switching to all-positive weights; positive diagonal similarity preserves each sign.

Status. Proven (conditional; see proof)

Next-step rationale. The sign degree of freedom must remain explicit; determine what a coarse observer loses.

04. ST8552 – Does the sign-cover automatically complete legacy into strict?

Outcome. Coarse mass is blind to every sign assignment, but its weights are $|\text{legacy}|$, not strict. The construction is not a completion map.

Status. Proven (conditional; see proof)

Next-step rationale. A mathematical dilation is insufficient: test the claimed dual dynamics at the level of vertex records.

05. ST8553 – Are heat and unitary vertex records the same process?

Outcome. Off-diagonal heat is $t W_{ij} + O(t^2)$; a finite Hamiltonian gives $t^2 |H_{ij}|^2 + O(t^3)$. No positive linear clock identifies them.

Status. Proven (conditional; see proof)

Next-step rationale. Try an explicit repeated-measurement limit instead of Wick continuation.

06. ST8554 – Does rapid projective measurement recover the strict heat operator?

Outcome. With H scaled as $\delta^{-1/2}$, repeated vertex measurements converge to rates $|H_{ij}|^2$. For $H=A$ these are W_{ij}^2 , not W_{ij} .

Status. Proven (conditional; see proof)

Next-step rationale. Try to compile the target rates explicitly, then test uniqueness of the compiled Hamiltonian.

07. ST8555 – Can one compile strict heat from a Hamiltonian, uniquely?

Outcome. Off-diagonal $H_{ij} = \sqrt{W_{ij}} \exp(i \theta_{ij})$ compiles W in that limit. Cycle phases and diagonals remain free and change coherent dynamics.

Status. Proven (conditional; see proof)

Next-step rationale. A compiled measurement limit introduces a Hamiltonian and instrument; test an exact quantum channel extension instead.

08. ST8556 – Is there an exact completely positive heat embedding?

Outcome. Rank-one jumps $\sqrt{W_{ij}} |j\rangle\langle i|$ give a trace-preserving Lindblad semigroup with exactly autonomous strict heat populations for all density matrices.

Status. Proven (conditional; see proof)

Next-step rationale. Exact embedding exists; test whether vertex records determine its coherence law.

09. ST8557 – Do complete heat population records select a quantum completion?

Outcome. Arbitrary nonnegative vertex dephasing changes coherences while preserving every autonomous population trajectory.

Status. Proven (conditional; see proof)

Next-step rationale. Test the desired original coherent generator $H=A$ in the same extension.

10. ST8558 – Can $H=A$ be added without altering exact autonomous heat populations?

Outcome. Not in the declared rank-one-jump plus vertex-dephasing class: autonomous heat for all density matrices forces H to be vertex-diagonal.

Status. Proven (conditional; see proof)

Next-step rationale. Do not identify two semigroups by a density-state Wick rotation; test normalization and linearity explicitly.

11. ST8559 – Does normalized imaginary-time filtering define a universal physical channel?

Outcome. It is nonlinear in the density matrix except for scalar filters. Unnormalized $P \rho P$ is a trace-decreasing CP operation, not deterministic heat on states.

Status. Proven (conditional; see proof)

Next-step rationale. Both stochastic and quantum lifts need extra premises. Test whether population consistency removes the stochastic freedom.

12. ST8560 – Does exchangeability plus exact projective consistency force independent walkers?

Outcome. No. Common vertex transpositions give a consistent exchangeable family with exactly the same single-particle generator Q and synchronized multi-particle jumps.

Status. Proven (conditional; see proof)

Next-step rationale. The fully synchronous law is reducible. Try irreducibility, detailed balance and the full equilibrium law as possible selectors.

13. ST8561 – Do irreducibility, reversibility and a product-uniform equilibrium select the kinetics?

Outcome. No. Every mixture $0 \leq \theta < 1$ is irreducible and reversible with the identical product-uniform equilibrium and exact singleton heat.

Status. Proven (conditional; see proof)

Next-step rationale. Static Shannon entropy is therefore insufficient. Locate a dynamical observable separating these realizations.

14. ST8562 – Which infinitesimal observation distinguishes the hidden common dynamics?

Outcome. Pair coincidence / cross quadratic variation distinguishes the generators even when every single-particle trajectory and equilibrium law agrees.

Status. Proven (conditional; see proof)

Next-step rationale. Ask exactly which extra dynamical premise makes singleton rates determine the full generator.

15. ST8563 – What minimal local premise yields a unique population generator?

Outcome. Autonomous singleton generator Q at every configuration plus no simultaneous coordinate changes forces $G_N = \sum_k Q^{\wedge}(k)$. Exchangeability is unnecessary.

Status. Proven (conditional; see proof)

Next-step rationale. Replace the qualitative no-simultaneous-jump premise by a measurable, nonnegative witness.

16. ST8564 – Can absence of common jumps be expressed without cancellation?

Outcome. The pair budget $B(x) = \sum_y g_{xy} \text{binom}(K(x,y), 2)$ is nonnegative and is zero exactly when every jump changes at most one coordinate.

Status. Proven (conditional; see proof)

Next-step rationale. A zero witness gives an exact theorem; derive a stable version for small nonzero coincidence rates.

17. ST8565 – Does approximate pair locality bound the full finite-time model error?

Outcome. With identical autonomous singleton rates, $\sup_x B(x) \leq \epsilon$ implies $\sup_x \text{TV}(P_{t(x, \cdot)}, P_{\text{ind}, t(x, \cdot)}) \leq \min(1, 2 \epsilon t)$.

Status. Proven (conditional; see proof)

Next-step rationale. Challenge the bound and its assumptions using many exchangeable particles with small jumps.

18. ST8566 – Do exchangeability and a deterministic large-population limit select shot noise?

Outcome. No. Uniformly chosen pairs with rate $\theta W/(N-1)$, mixed with single jumps, have the same mean heat and $O(1/N)$ empirical variance but an altered noise prefactor.

Status. Proven (conditional; see proof)

Next-step rationale. This family is not projectively consistent across N . Test whether moment data can certify the missing event structure directly.

19. ST8567 – Which raw-event inequalities survive arbitrary nonnegative jump rates?

Outcome. Local integer-count cumulants obey $\kappa_2 \geq |\kappa_1|$ and $\kappa_4 \geq \kappa_2$; equality in the latter excludes all marks with $|k| \geq 2$.

Status. Proven (conditional; see proof)

Next-step rationale. These are instantaneous raw-count statements. Attempt to falsify their use with ordinary detector binning.

20. ST8568 – Can an ordinary detector make valid Poisson events appear to violate the local count bound?

Outcome. Yes. A one-click-per-bin threshold maps Poisson arrivals to Bernoulli clicks, with variance/mean $= \exp(-\lambda \Delta) < 1$. Finite-bin data cannot be inserted into the instantaneous theorem.

Status. Proven (conditional; see proof)

Next-step rationale. Separate removable loss from irreversible thresholding and test whether a robust coincidence protocol exists.

21. ST8569 – Which detector imperfection preserves the zero-coincidence theorem?

Outcome. Independent resolution-preserving Bernoulli thinning gives $B_{\text{observed}} = \eta^2 B$. For $\eta > 0$ it preserves the exact zero set; a quantitative bound needs a lower efficiency bound.

Status. Proven (conditional; see proof)

Next-step rationale. Finite bins mix multiple events. Bound that contamination instead of assuming it vanishes.

22. ST8570 – Can finite time resolution still test absence of simultaneous changes?

Outcome. With a finite escape bound Λ , a single-change null has $\Pr(\text{at least two detected changes}) \leq (\Lambda \Delta)^2/2$. A resolved joint event gives a positive order- Δ lower bound.

Status. Proven (conditional; see proof)

Next-step rationale. The rate orders separate analytically; compute a finite-sample test without synthetic-data fitting.

23. ST8571 – Does the proposed rate-order test have a finite resource specification?

Outcome. For independently reset trials a predeclared binomial upper-tail test controls the null and has high calculated power in the supplied two-particle model. No physical trial was performed.

Status. Proven conditional test rule; numerical tails supplemented by exact rational bounds

Next-step rationale. Attempt to destroy identifiability with an uncalibrated but nondegenerate detector.

24. ST8572 – Can unknown event merging exactly hide common dynamics?

Outcome. Yes. One-click-per-event reporting of the common-flip mixture and independently thinned single flips can produce the same entire nonzero Poisson record law despite different microscopic generators.

Status. Proven (conditional; see proof)

Next-step rationale. Even a mark-preserving detector may leave a clock/efficiency ambiguity. Test the full record likelihood, not just its mean.

25. ST8573 – Does full resolved-count likelihood determine both clock and efficiency?

Outcome. No. The two-particle count law has only c_1, c_2 ; a simultaneous change of w, θ, η preserves both and hence every counting distribution.

Status. Proven (conditional; see proof)

Next-step rationale. Operational nonuniqueness is explicit. Audit the earlier nonlinear/prism and inverse-action alternatives before the final synthesis.

26. ST8574 – Can the failed pure-parity kinetic counterexample be repaired honestly?

Outcome. Adding an explicitly declared symmetric field gives two flip-energy magnitudes, so Metropolis and heat-bath cease to differ only by a clock. The original pure-parity proportionality is retained.

Status. Proven (conditional; see proof)

Next-step rationale. Equilibrium geometry fails to select kinetics. Test the analogous inverse-propagator-to-action claim.

27. ST8575 – Does a supplied propagator determine the nonlinear variational theory?

Outcome. No. Every $S_{\lambda} = \phi^T M \phi / 2 - J^T \phi + \lambda \sum \phi_i^4$, $\lambda \geq 0$, has the identical zero-source Hessian and linear response M^{-1} but different cubic response.

Status. Proven (conditional; see proof)

Next-step rationale. Test whether hidden-state compression can select the missing dynamical response from the same static Green data.

28. ST8576 – Does exact static Green/Schur compression select a hidden memory law?

Outcome. No. Positive two-variable actions have identical coarse static Green function but different rational response and exponential memory; iteration does not remove the unsourced hidden parameters.

Status. Proven (conditional; see proof)

Next-step rationale. The surviving ambiguities occur in signs, quantum instruments, collective jumps, interactions and memory. Formulate only the no-go actually supported by explicit models.

29. ST8577 – Can the audited finite FIN data logically determine a unique collective physical realization?

Outcome. Not in the stated finite Markov completion class: same W , all singleton paths, product equilibrium, detailed balance, projective consistency and FIN graph symmetries coexist with different pair records.

Status. Proven (conditional; see proof)

Next-step rationale. This is not a universal impossibility of future physics. Close with a premise-removal and robustness audit of the smallest demonstrated positive bridge.

30. ST8578 – Which rigorously narrowed bridge survives premise removal and perturbations?

Outcome. In a declared finite CTMC, autonomous singleton Q plus zero pair-coincidence budget uniquely determines the generator; a small budget gives a stable TV bound. Initial law, clock, detector and the source of zero budget remain separate obligations.

Status. Proven (conditional; see proof)

Next-step rationale. Next: seek a strict-derived composition law forcing the nonnegative pair budget to vanish, or prove its independence in a precisely declared enlarged source class.